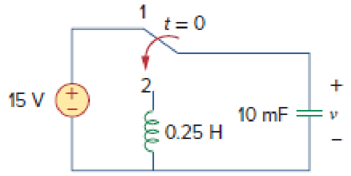
Chapter 16, Problem 64P

Problem

The switch in Fig. 16.87 moves from position 1 to position 2 at $t = 0$. Find $v(t)$, for all $t > 0$.

Figure 16.87:



Step-by-step solution

Step 1 of 2

Capacitor at t = 0+ behaves like a short circuit and at t = ∞ it behaves like an open circuit.

Inductor at t = 0+ acts like an open circuit and at t = ∞ acts like a short circuit.

Infinite amount of time switch is at position “1” capacitor voltage and inductor current cannot change instantaneously.

$$v(0^-) = v(0^+) = 15 \text{ V}$$
$$i_L(0^-) = i_L(0^+) = 0 \text{ A}$$

With initial values the switch is at position “2” at t = 0 sec.

Step 2 of 2

Apply the Kirchhoff's voltage law.

$$\frac{-15}{s}-\frac{I(s)}{s(0.01)}-0.25sI(s)=0$$

$$I(s)=\frac{-(15/s)}{\left[\frac{100}{s}+0.25s\right]}$$
$$=\frac{-15}{\left[100+0.25s^2\right]}$$

Determine the expression for the voltage $v(s)$.

$$v(s)=\frac{15}{s}+\frac{I(s)}{s(0.01)}$$
$$=\frac{15}{s}+\frac{(-15)100}{s\left(100+0.25s^2\right)}$$
$$=\frac{15}{s}-\frac{1500}{s(0.25)\left(s^2+\frac{100}{0.25}\right)}$$
$$=\frac{15}{s}-\frac{6000}{s\left(s^2+400\right)}$$
$$=\frac{(60/8)}{s-j20}+\frac{(60/8)}{s+j20}$$
$$=\frac{7.5}{s-j20}+\frac{7.5}{s+j20}$$

Applying inverse Laplace transform.

$$v(t)=7.5\left(\frac{e^{j20t}+e^{-j20t}}{2}\right)\times 2u(t)$$
$$=15\cos(20t)u(t)\text{ V}$$

Hence the voltage across the capacitor is $\boxed{15\cos(20t)u(t)\text{ V}}$.